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Reference-free Detection of LSB Steganography Using Histogram Analysis

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***Abstract*— Due to the difficulty of obtaining a database of original images that are required in the classification process to detect tampering, this paper presents a technique for detecting image tampering such as image steganography in the spatial domain. The system depends on deriving the auto-correlation function of the image histogram, then applying a high-pass filter with a threshold. This technique can be used to decide which image is cover or a stego image, without adopting the original image. The results have eventually revealed the validity of this system. Although this study has focused on least-significant-bit (LSB) steganography, we expect that it could be extended to other types of image tapering.**

***Keywords*— steganography, steganalysis, LSB steganography, image tampering.**

I. Introduction

The continuous innovation propels the individuals to share increasingly more data among each other. For instance, there are few associations like medicine, military institutes that are offering information which are exceptionally cryptic and significant. Concerning the secure messaging, the individuals utilize cryptography which has the secret key aiming that the recipient can unscramble the message and verify that the message stays unblemished. However, the cryptography raised doubt among aggressors and their attempt to assault the message to get the shrouded messages. In this way, we should adopt a new method of steganography which includes an innocent cover message embedded with secret message optionally scrambled so that while moving least doubt excite among aggressors. In any case, in the event that this methodology is utilized by criminal association, at that point it gets necessary to distinguish the stego multimedia data and attempt to get the information installed into it. So, similar to cryptanalysis which works on cryptography, steganalysis is an art of dissuading covert communication without influencing the honest ones. The main role of steganalysis is to identify the covert message and whenever expected, attempt to discover more information like secret message length, steganography strategy utilized, etc. [1].

II. Steganalysis Techniques to Detect LSB Steganography

Steganalysis is a specialty of identifying the hidden messages inside a stego medium. Steganalysis can be passive or active. In passive steganalysis, the attacker attempts to decide if the medium is a stego medium. In active steganalysis, the attacker attacks the stego medium, distinguish and alter the stego messages. The presence of embedded messages is frequently intangible; however, it may disturb the statistics of an image. Identifying the matter of some statistical characters between the cover and stego media becomes key issue in steganalysis [2]. There are different steganalysis methods [3].

Steganalysis algorithms are classified as follows:

1. Specific steganalysis: These types of procedures are formed by analyzing the embedding activity and deciding certain image statistics. Such systems need a detailed knowledge of embedding process. These strategies can achieve very accurate results when utilized

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against an objective steganography system. Specific statistical steganalysis apparatuses are utilized for discovering secret message from stego-images embedded by LSB embedding. The disadvantage of using this strategy is that it is extremely limited to specific embedding algorithm and the image format.

2. Blind /Universal steganalysis: Universal statistical steganalysis involves the statistical steganalysis technique that isn't custom-fitted for a particular steganography embedding strategy. It needs less or even no prior information of the under-attack steganographic techniques used for discovery of secret message.

III. Related Works

LSB substitution steganography is the least difficult addition strategy. There are great steganalysis strategies for detecting hidden message. Some of them are in the spatial domain and the absolute rest are in transform domain. Some of them attack specifically LSB embedding technique and others are universal methods that attack LSB strategy. Also, some steganalysis strategies can realize stego-images, while some others can distinguish inclusion rate as well. Steganalysis techniques for LSB substitution steganography are divided in two groups: the first group works just on color images or only on gray-scale images [4,5]; while the second group works on both color and gray-scale images [6,7].

The RS steganalysis method is a viable steganalytic system suggested by Fridrich et al. [6]. The RS steganalysis strategy is solid even in the identification of non-sequential LSB embedding in digital images.

Andrew D. Ker [7] set a general frame work for structural analysis of LSB substitution discovery and furthermore message length evaluation. They united some previously known structural detectors and displayed an all-the-more controlled technique. In addition, they use different image in JPEG and bmp format to check other suggested strategies and compare it with some different techniques; they likewise revealed that new algorithm has high rank performance in a bigger number of situations when message length is over 2%.

Tao Zhang and Xijian Ping, suggested a steganalysis strategy for realization of LSB steganography in natural images dependent on difference image histogram, and explained that this algorithm can dependably identify steganography and evaluate embedded message rate. This technique doesn't have a dependable execution for low addition rates [8].

Fridrich et al. [6] proposed a steganalysis method dependent on image's biplanes correlation and pointed out that LSB plane can be evaluated from the other seven ones. Be that as it may, when the LSB plane's matter is further randomized it has not a solid execution.

Kong et al. [9] explained a steganalysis algorithm that is dependent on the way of hiding secret bits in LSB plane and randomizing them. Mehmet U. Celik et al, [10] suggested another steganalysis algorithm dependent on rate-distortion, which depends on the way that steganographic change signal statistics, for example, rate-distortion characteristic for signal (image). At that point they exhibited that this algorithm has a good presentation on LSB substitution steganography technique. Writer uses just Kodak database with high quality images for checking their method. Their technique doesn't have a decent presentation for low insertion rate.

Gurong Xuan et al [11] proposed a steganalysis strategy that is dependent on various features extracted from statistical moments of wavelet functions capacities. They picked a 39-D feature vector of images in wavelet domain as recognition function. They compare the suggested algorithm with two previous algorithms and explained that new method has a high-rank execution.

Avcibas et al [12] used three types of recognizing features for classifying embedded and non-embedded images. They used features like: similarity differences, histogram and entropy-related features, and a lot of measures relying on a neighborhood-weighting mask in spatial domain. In fact, authors used correlation between the contiguous bit planes, specially seven- and eight-bit plane, as a criterion for recognizing embedded image from nature image.

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IV. SUITABLE IMAGES FOR LSB STEGANOGRAPHY

Human visual framework (HVS) isn't delicate to extremely little color variations. LSB steganography strategies took advantage of this human eye's shortcoming in recognizing little detail varieties, and insert message information in least significant bits as noise. Since channel capacity constraint is a major test in transmission systems, different compression methods are used to minimize the measure of information that will be sent. These techniques took advantage of human visual system limitation as well, similar to LSB steganography strategy, and delete redundant pieces of information that will transmit (like image) [7]. For this reason, compressed image like jpeg for the most part are bad decisions for steganography of LSB in spatial space (because of compression, most of its redundant information is deleted). Be that as it may, for inserting in transform domain, as DCT, secret bits can be inserted in LSB bits of DCT coefficients. These sorts of steganographic strategies are strong yet low capacity. On the other hand, the uncompressed images have redundant information that are suitable for LSB insertion in spatial domain (more capacity for hiding secret message than compressed image), without making any genuine perceptual change in image. This makes steganalysis to be more confused against compressed stego-image than uncompressed stego-image.

V. CORRELATION COEFFICIENTS

Correlation as a general concept is a measure of an association between variables. In correlated data, the change in the magnitude of one variable is related with an adjustment in the extent of another variable, either in the equivalent (positive correlation) or in the inverse (negative correlation) direction. Frequently, the term correlation is used in the context of a linear relationship between 2 continuous variables and expressed as Pearson product-moment correlation.

The Pearson correlation coefficient is commonly used for jointly normally distributed data (data that follow a bivariate normal distribution). For nonnormally appropriated continuous data, for ordinal data, or for data with significant outliers, a Spearman rank correlation can be utilized as a measure of a monotonic association. Both correlation coefficients are scaled with the end goal that they run from – 1 to +1, where 0 demonstrates that there is no linear or monotonic association, and the relationship gets stronger and at last methods a straight line (Pearson correlation) or an always increasing or decreasing curve (Spearman correlation) as the coefficient approaches an absolute value of 1. Hypothesis tests and certainty intervals can be utilized to address the statistical significance of the outcomes and to estimate the strength of the relationship in the population from which the data were sampled [13].

VI. AUTOCORRELATION FUNCTION (ACF)

The autocorrelation function (ACF) is a generally known measurement for identifying the presence of serial correlation. The ACF is more helpful than the tests referenced above, as it gives an increasingly point by point depiction of the underlying procedure. For a real signal x(n) with length N samples, the autocorrelation is calculated as [14]:

$$R_x(k) = \frac{1}{N}\sum_{n=0}^{N} x(n)x(n+k) \ldots (1)$$

VII. THE PROPOSED SYSTEM

According to the correlative analysis explained above, it is suggested to use the extended system of LSB steganalysis method. LSB steganalysis is a method for detecting the presence of secret message in the cover image. The proposed system computes the autocorrelation function of the image histogram, then takes the first or second derivatives of the autocorrelation function to test whether any steganography exists in the image. If LSB steganography exists, then the derivatives of the autocorrelation function will have significant ripples, and a high-pass filter can extract the high-pass content, where a threshold could be used for a decision.

VIII. RESULTS AND DISCUSSION

The study tested the proposed system using LSB steganography with 1, 2 and 4 LSB bits to hide message in the cover image, where a grey-scale image was used. The approach should discover whether this image is a cover or a stego. Figure 1 shows the cover and the stego message used in this

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experiment, where the message has been re-sized to get different message-to-cover ratios (Rm), while Figure 2 shows the normalized autocorrelation of the histograms of the cover and stego images using k-LSBs steganography system (i.e., the stego image, the cover image loaded with a hidden message as another image whose pixels have been distributing over k-LSB's of the cover image).

The embedding of a message inside the cover image has disturbed the autocorrelation curves of the cover image, where the original curve has changed from a nearly smooth curve to curves with ripples. This damage is not clearly evident in these curves, nor in the histograms of these images. However, the proposed system shows a way to clarify the changes and reveal tampering (with a message as an image) undoubtedly as one can see from Figure 3.

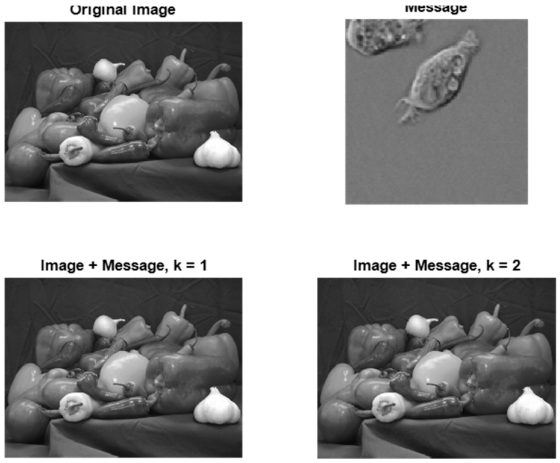

Figure 1: The cover and stego MATLAB images.

Figure 3 shows the effect of the first derivative on the histogram autocorrelation of the cover and LSB-stego images. It can be noticed that LSB-stego will introduce significant ripples when the autocorrelation function of the image histogram is subjected to derivation. Hence, a high-pass filtering of this derivative will be a good tool to discover these ripples, where a threshold should be used as will be explained below.

On the other hand, it is found that using higher derivatives may not be necessary, as they do not introduce a significant difference from the first derivative in revealing image tampering. Figures 4 and 5 show the ripples introduced by the second and third derivatives of the histogram autocorrelation functions.

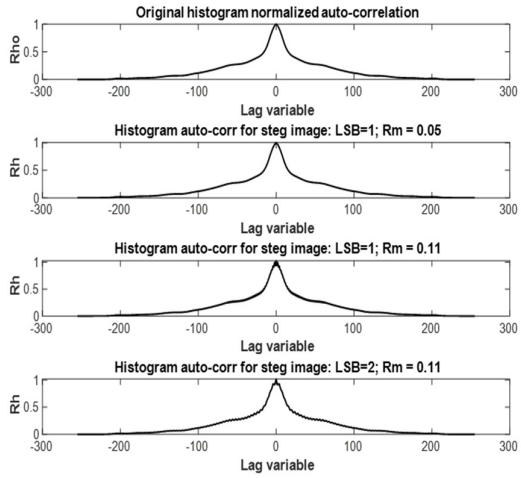

Figure 2: Normalized autocorrelation of the histograms of the cover and stego-image (cover image loaded with a message) using k-LSBs steganography system.

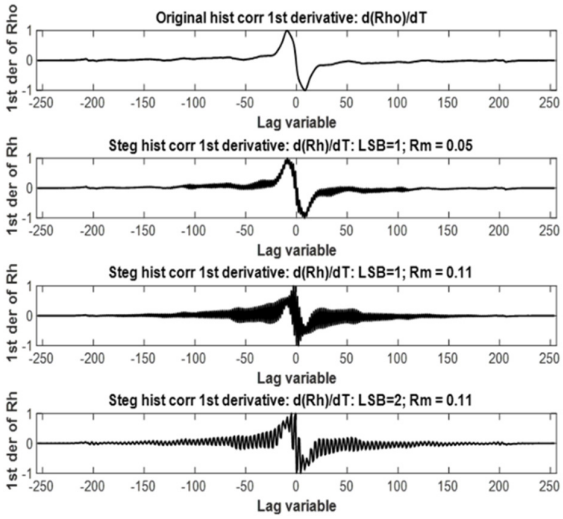

Figure 3: First derivative of the histogram auto-correlation function of the cover and LSB-stego images for different LSB-levels and different message to cover size ratios.

Hence, the second and third derivatives can be used as a support for the decision which can be based on the first derivative only. However, this comment may not be

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considered as a final conclusion on the role of higher derivatives as the Authors haven't fully studied their effect yet.

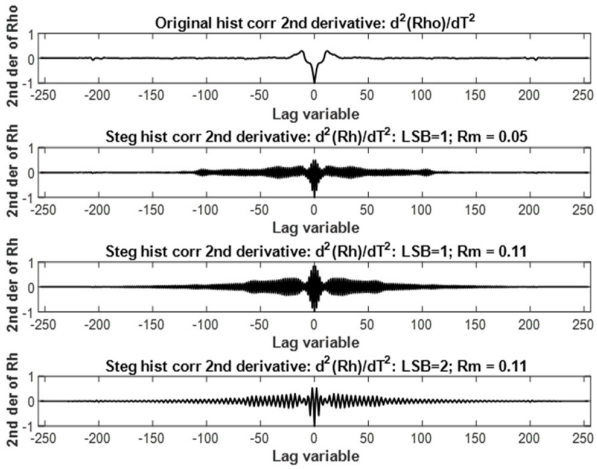

Figure 4: The second derivative of the histogram auto-correlation function of the cover and LSB-stego images for different LSB-levels and different message to cover size ratios. Please compare with Figures 2, 3.

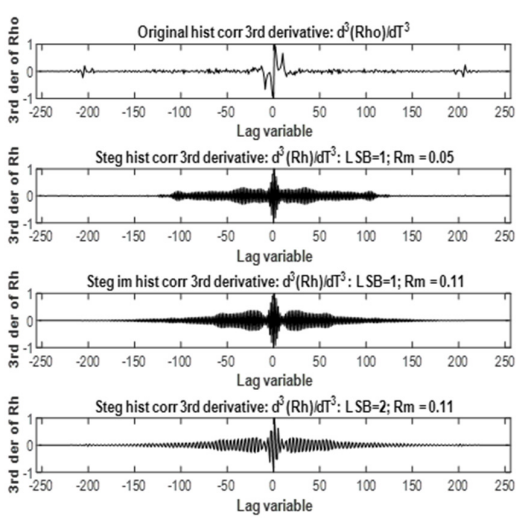

Figure 5: The third derivative of the histogram auto-correlation function of the cover and LSB-stego images for different LSB-levels and different message to cover size ratios. Please compare with Figures 3 and 4.

It worth noting that the pixel-to-pixel correlation decreases with the ratio Rm, which is the message size to the cover image size. Figure 6 confirms this fact. In order to find the correlation between two adjacent pixels (horizontal, vertical and diagonal), 5000 pairs of adjacent pixels have been selected randomly from the original and encrypted images. Figure 6 shows the correlation of adjacent pixels in the cover and k-LSB-stego images for different values of k (the number of least significant bits to be encoded with the message).

It is clear that introducing a message in the cover image will reduce the natural correlation between its pixels. The correlation damage will be more if we used more LSB's (i.e., larger k). Hence, for better security of the message, the message size should be selected as small as possible when compared to the cover image.

The correlation analysis is calculated according to the Equation (2) below, and the range of the coefficient is [-1,1]:

$$R_{xy} = \frac{\mathrm{cov}(x,y)}{\sqrt{D(x)} \times \sqrt{D(y)}} \tag{2}$$

where:

$$\mathrm{cov}(x,y) = \varepsilon\big(x - \varepsilon(x)\big)\big(y - \varepsilon(y)\big) \tag{3}$$

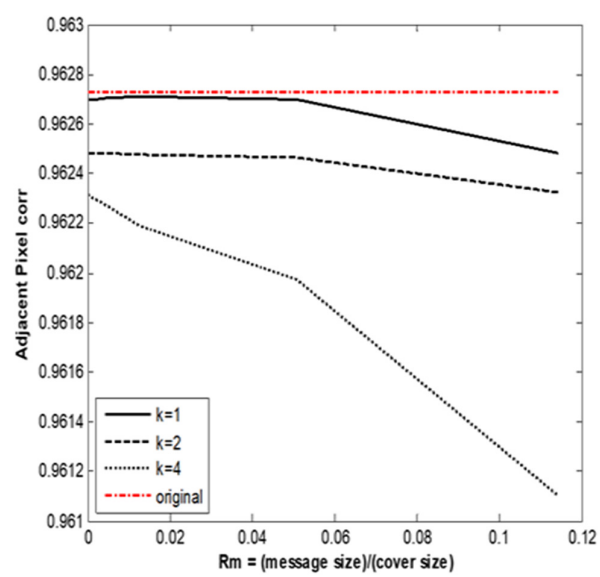

Figure 6: Correlation of adjacent pixels in the cover and k-LSB-stego images for different k.

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$$\varepsilon(x) = \frac{1}{N}\sum_{i=1}^{N}(x_i) \quad (4)$$

$$D(x) = \frac{1}{N}\sum_{i=1}^{N}(x_i - \varepsilon(x))^2 \quad (5)$$

$$\mathrm{cov}(x,y) = \frac{1}{N}\sum_{i=1}^{N}\big(x_i - \varepsilon(x)\big)\big(y_i - \varepsilon(y)\big) \quad (6)$$

The high-pass filter used in this research is digital Butterworth filter of order 10 using the following MATLAB instructions:

```
n=10; wn=0.5;
[b,a] = butter(n,wn,'high');
```

and whose frequency response is shown in Figure 7.

After HPF of the derivative of the autocorrelation function of the histogram, the power of the filtered signal is calculated. As the original image is smoothly correlated with smooth histogram autocorrelation, the filtered version of the derivative of histogram autocorrelation will have very low paper.

However, if the image is tampered with, then the power of the filtered derivative of the histogram-autocorrelation will be high.

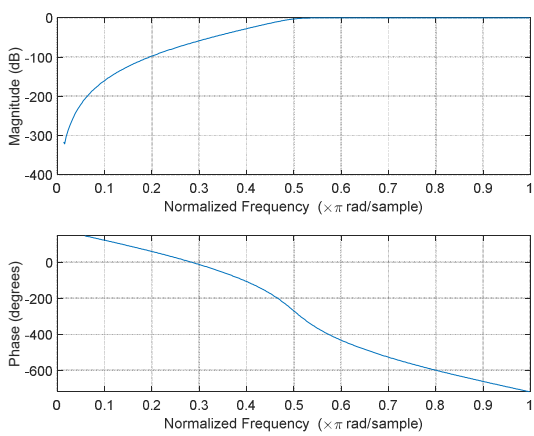

Figure 7: The frequency response of the high-pass filter used to determine the threshold of tampering detection.

In the above experiments we found that the ratio of the tampered power to the original power is more evident if we use the first derivative, where this ratio exceeds 1000. For untampered natural image, the power of its filtered derivative of histogram-autocorrelation was as low as 0.0015, while the power of the tampered one was 2.3733; hence a threshold of 0.1 or higher would be sufficient to detect tampering, although the threshold could be image-dependent. More studies are expected to define a closer estimation of the threshold.

IX. Conclusion

A system for steganalysis in the spatial domain is proposed. This system is tested for LSB steganalysis, but it could be extended to other steganographic approaches. The method utilizes correlative analysis (applied to the image histogram) to detect which image is cover or stego. The proposed system can be used to detect the tampering in images without adopting the original image. The method is based on the fact that the derivation of the autocorrelation function of the image histogram would exhibit significant ripples if the image has been tampered with a stego message. Hence, high-pass filtering of the histogram autocorrelation function would reveal steganography when a proper threshold is applied to the power of the filtered derivative of the image histogram-autocorrelation.

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